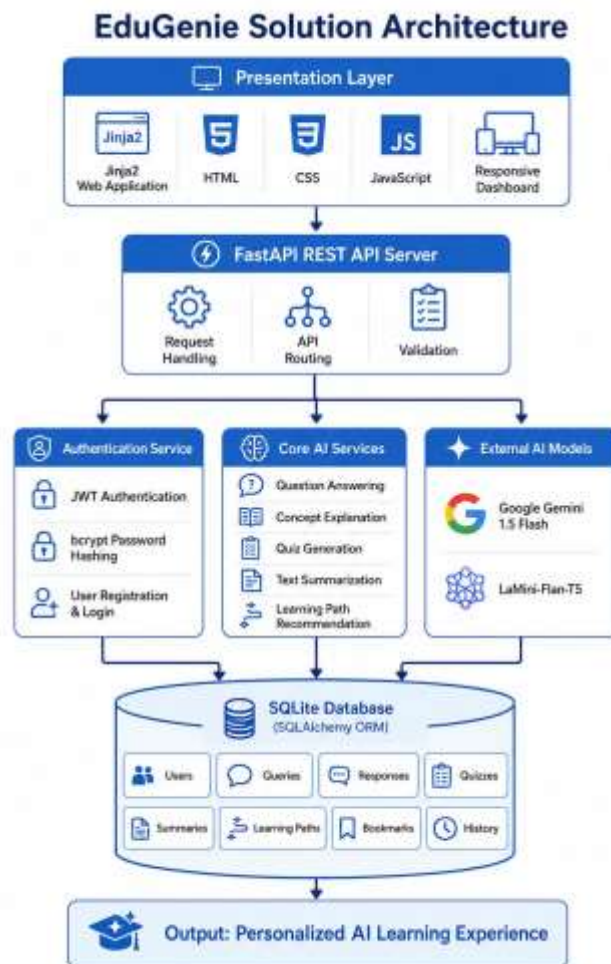


## Solution Architecture

|                      |                                                   |
|----------------------|---------------------------------------------------|
| <b>Date</b>          | 28 June 2026                                      |
| <b>Team ID</b>       | SWTID-2026-3658                                   |
| <b>Project Name</b>  | EduGenie Google Gemini Powered Learning Assistant |
| <b>Maximum Marks</b> | 5 Marks                                           |

### Solution Architecture Diagram:

Provide a high-level visual representation of your solution's architecture. Use the structural layout below as a template to map out your Presentation Layer, Application Layer, and Data Layer. Fill in the specific technologies and components for your project directly into the boxes to create your architectural diagram.



**Component Description Table**

| Component Name           | Description / Role in Architecture                                                                                                                                  | Technologies Used                       |
|--------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------|
| Presentation Layer       | Provides a responsive user interface for login, question answering, concept explanation, quiz generation, summarization, and personalized learning recommendations. | Jinja2, HTML, CSS, JavaScript           |
| API Gateway              | Handles incoming API requests, validates user input, routes requests to AI services, and returns JSON responses.                                                    | FastAPI, Uvicorn                        |
| Core Logic Service       | Processes educational queries, coordinates AI models, generates quizzes, summaries, explanations, and learning paths.                                               | Python, SQLAlchemy                      |
| Authentication Service   | Manages user registration, login, password hashing, JWT token generation, and secure session handling.                                                              | bcrypt, JWT, Python                     |
| Database / Storage       | Stores user profiles, AI responses, quizzes, summaries, learning history, bookmarks, and learning paths.                                                            | SQLite, SQLAlchemy                      |
| External API Integration | Provides AI-powered educational assistance using Google Gemini for reasoning tasks and LaMini-Flan-T5 for concept explanations.                                     | Google Gemini 1.5 Flash, LaMini-Flan-T5 |
| Testing Layer            | Verifies API endpoints, database operations, authentication, and AI service integration.                                                                            | PyTest, FastAPI TestClient              |